Hexadigitall Technologies

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SCAN FOR COURSE

CORE TRACK - PREREQUISITES REQUIRED

DevOps Engineering & Cloud Infrastructure

DURATION
20 Weeks

LEVEL
Advanced

SCHOOL
Cloud & DevOps

Welcome to Core DevOps Engineering

This intensive 20-week **Core Track** focuses exclusively on modern cloud infrastructure and DevOps practices: Docker containerization, Kubernetes orchestration, Terraform Infrastructure as Code, CI/CD automation, GitOps workflows, and production monitoring systems.

Unlike our comprehensive 20-week program, this accelerated track assumes you already have foundational knowledge in Linux, Git workflows, scripting, and configuration management. We dive straight into containerization, orchestration, and cloud infrastructure engineering.

⚠️ Important: Prerequisites Required

This is a **focused, advanced curriculum** that does NOT cover foundational topics in depth. To succeed, you must already be proficient in:

→ **Linux Administration:** Command-line operations, file systems, permissions, process management

→ **Version Control:** Git workflows, branching, merging, and repository management

→ **Scripting:** Python or Bash for automation tasks

→ **Configuration Management:** Understanding of Ansible or similar tools

Need these skills? Enroll in our prerequisite courses first (see Required Foundation Courses section below).

🎓 Required Foundation Courses

Before enrolling in this Core Track, you **must complete** or demonstrate proficiency in these areas. We offer dedicated courses for each:

🐧 Linux Administration

Required Course:
Linux Administration & Shell Scripting Pro

You'll need solid Linux skills including command-line mastery, system administration, shell scripting, and server management. This course is essential for DevOps work.

🔀 Version Control

Required Course:
Git & GitHub for Beginners

Git is the backbone of DevOps workflows. You must understand branching strategies, merge conflicts, pull requests, and collaboration workflows before starting this course.

🌀 Scripting & Automation

Recommended Course:
Python for Data Science & Analytics

While we cover some scripting basics, having Python knowledge accelerates your learning. Python is widely used for DevOps automation and infrastructure tooling.

⚙️ Configuration Management

Recommended Course:
Advanced Ansible Automation & IaC

This Core Track touches on Ansible briefly. For deep configuration management expertise, take our dedicated Ansible course which covers playbooks, roles, and enterprise automation.

💡 Course Bundle Recommendation

For the best learning experience, we recommend taking these courses in this order:

1. **Git & GitHub for Beginners** (4 weeks)
2. **Linux Administration & Shell Scripting Pro** (10 weeks)
3. **Python for Data Science & Analytics** (12 weeks) - Optional but highly beneficial
4. **DevOps Engineering & Cloud Infrastructure (Core Track)** (14 weeks) - This course!
5. **Advanced Ansible Automation & IaC** (20 weeks) - For advanced configuration management

What You'll Master in 20 Weeks

This Core Track is laser-focused on essential DevOps tools and practices:

→ **Docker & Containerization:** Build, optimize, and manage containerized applications at scale

→ **Kubernetes Orchestration:** Deploy and manage production-grade Kubernetes clusters

→ **Infrastructure as Code:** Provision cloud resources with Terraform across AWS and Azure

→ **CI/CD Pipelines:** Build automated deployment pipelines with Jenkins, GitHub Actions, and GitLab

→ **GitOps Workflows:** Implement declarative deployments with ArgoCD and Flux

→ **Monitoring & Observability:** Set up Prometheus, Grafana, and ELK stack for production systems

→ **Cloud Platform Mastery:** Hands-on experience with AWS and Azure services

→ **DevSecOps Practices:** Security scanning, secrets management, and compliance automation

Getting Started

What You'll Need:

→ Computer with 16GB RAM minimum (for running VMs, Docker, and Kubernetes locally)

→ AWS Free Tier account (we'll guide setup in Week 1)

→ GitHub account with existing Git knowledge

→ Docker Desktop installed (instructions provided)

→ Proficiency in Linux command-line (prerequisite course required)

→ Basic scripting ability in Python or Bash (prerequisite recommended)

Advanced Specialization Tracks

After completing this Core DevOps course, deepen your expertise with these advanced programs:

→ **AWS Certified Solutions Architect** - Master AWS cloud architecture and prepare for certification

→ **Kubernetes Quick Start** - Deep dive into advanced Kubernetes patterns and production operations

→ **DevSecOps Engineering: Automating Security -** Integrate security throughout the DevOps lifecycle

→ **Advanced Ansible Automation & IaC** - Enterprise configuration management and automation

→ **Cloud Infrastructure Deployment Strategy** - Multi-cloud architecture and deployment patterns

Learning Resources

📘 Free Resources

- → Docker Official Documentation
- → Kubernetes Documentation (kubernetes.io)
- → Terraform Registry & Docs
- → AWS Free Tier & Documentation
- → GitHub Actions Documentation
- → Prometheus & Grafana Tutorials
- → CNCF Landscape & Project Docs

📚 Essential Books

- → Kubernetes in Action (Marko Lukša)
- → Terraform: Up & Running (Yevgeniy Brikman)
- → Site Reliability Engineering (Google)
- → The DevOps Handbook (Gene Kim)
- → Continuous Delivery (Jez Humble)

🛠️ Core Tools

- → Docker Desktop
- → Kubernetes (minikube/kind)
- → Terraform CLI
- → AWS CLI & Console
- → GitHub Actions / Jenkins
- → ArgoCD
- → Prometheus & Grafana

🎓 Practice Platforms

- → Play with Docker
- → Kubernetes Playground
- → AWS Free Tier Sandbox
- → Terraform Cloud (free tier)
- → GitHub (free private repos)

20-Week Core Track Roadmap

This program assumes foundational knowledge and focuses on core DevOps technologies:

→ **Weeks 1-2:** DevOps Overview & Environment Setup - CI/CD concepts, Docker fundamentals, containerization

→ **Weeks 3-5:** Advanced Docker & Containerization - Multi-container apps, Docker Compose, security, optimization

→ **Weeks 6-10:** Kubernetes Mastery - Architecture, deployments, networking, storage, Helm, production patterns

→ **Weeks 11-13:** Infrastructure as Code with Terraform - AWS/Azure provisioning, modules, state management

→ **Weeks 14-16:** Advanced CI/CD & GitOps - Pipeline automation, GitOps workflows, deployment strategies

→ **Weeks 17-19:** Monitoring, Security & Production Operations - Observability, DevSecOps, SRE practices

→ **Week 20:** Capstone Project - End-to-end platform implementation and presentation

📝 Note on Course Focus

This curriculum intentionally minimizes coverage of:

→ **Linux basics** - We assume command-line proficiency from prerequisite course

→ **Git workflows** - We assume Git knowledge; only cover advanced CI/CD integration

→ **Python scripting** - We focus on tools, not language fundamentals

→ **Ansible deep-dive** - Briefly covered; dedicated course available for mastery

This focused approach allows us to go deeper on containerization, orchestration, and cloud infrastructure in a shorter timeframe.

Weekly Curriculum Breakdown

1 DevOps Foundations & CI/CD Overview

TOPICS COVERED

- → DevOps culture and lifecycle
- → Continuous Integration/Delivery principles
- → Deployment strategies overview
- → Cloud platforms introduction (AWS/Azure)
- → Modern infrastructure patterns
- → Environment setup and tooling

HANDS-ON LABS

- → AWS account setup and configuration
- → Development environment preparation
- → Simple CI pipeline demonstration
- → Tool installation and verification

Prerequisite Check: We briefly review Git and Linux essentials. If you struggle here, pause and complete the prerequisite courses.

2 Docker Fundamentals & Containerization

TOPICS COVERED

- → Container technology deep dive
- → Docker architecture and components
- → Images, containers, and layers
- → Dockerfile best practices
- → Container networking fundamentals
- → Volume management and persistence

HANDS-ON LABS

- → Install and configure Docker
- → Build custom Docker images
- → Container lifecycle management
- → Networking and storage setup

3 Advanced Docker & Multi-Container Applications

TOPICS COVERED

- → Docker Compose for orchestration
- → Multi-stage builds and optimization
- → Container security best practices
- → Docker registry and image management
- → Health checks and restart policies
- → Resource limits and constraints

HANDS-ON LABS

- → Multi-tier app with Docker Compose
- → Image optimization techniques
- → Security scanning with Trivy
- → Push images to container registry

4 Container CI/CD & Automation

TOPICS COVERED

- → Automated container builds in CI
- → Container testing strategies
- → Image scanning in pipelines
- → Automated deployment workflows
- → GitHub Actions for containers
- → Registry integration patterns

HANDS-ON LABS

- → Build container CI/CD pipeline
- → Automated testing in containers
- → Security scanning integration
- → Deploy to cloud container service

5 Kubernetes Architecture & Core Concepts

TOPICS COVERED

- → Kubernetes architecture overview
- → Control plane and worker nodes
- → Pods, ReplicaSets, Deployments
- → Services and networking model
- → kubectl command-line mastery
- → Local cluster setup (minikube/kind)

HANDS-ON LABS

- → Install local Kubernetes cluster
- → Deploy first application
- → Create deployments and services
- → Explore cluster components

6 Kubernetes Workloads & Configuration

TOPICS COVERED

- → Namespaces and resource isolation
- → ConfigMaps and Secrets
- → Labels, selectors, and annotations
- → Resource requests and limits
- → Environment configuration patterns
- → Multi-environment management

HANDS-ON LABS

- → Namespace isolation setup
- → ConfigMaps and Secrets management
- → Resource quota implementation
- → Multi-environment deployment

7 Kubernetes Networking & Storage

TOPICS COVERED

- → Kubernetes networking model
- → Ingress controllers and rules
- → Network policies and security
- → Persistent Volumes and Claims
- → Storage classes and provisioning
- → StatefulSets for stateful apps

HANDS-ON LABS

- → Configure Ingress controller
- → Implement network policies
- → Set up persistent storage
- → Deploy stateful application

8 Advanced Kubernetes Patterns

TOPICS COVERED

- → DaemonSets, Jobs, and CronJobs
- → Init containers and sidecars
- → Horizontal Pod Autoscaling
- → Rolling updates and rollbacks
- → Health checks (liveness/readiness)
- → Pod disruption budgets

HANDS-ON LABS

- → Implement autoscaling
- → Configure health checks
- → Rolling update strategies
- → Schedule jobs and cron tasks

9 Helm & Kubernetes Package Management

TOPICS COVERED

- → Helm architecture and concepts
- → Charts, templates, and values
- → Helm repositories and artifact hub
- → Creating custom charts
- → Release management
- → Helm best practices

HANDS-ON LABS

- → Install apps with Helm
- → Create custom Helm charts
- → Template complex deployments
- → Manage chart dependencies

10 Infrastructure as Code with Terraform

TOPICS COVERED

- → IaC principles and Terraform workflow
- → HCL syntax and configuration
- → Providers, resources, data sources
- → Variables, outputs, and locals
- → State management fundamentals
- → AWS resource provisioning

HANDS-ON LABS

- → Write Terraform configurations
- → Provision AWS infrastructure
- → Manage state files
- → Use variables and outputs

11 Advanced Terraform & Cloud Provisioning

TOPICS COVERED

- → Modules and code reusability
- → Remote state with S3 and locking
- → Workspaces for environments
- → Provisioning Kubernetes (EKS/AKS)
- → Terraform Cloud and collaboration
- → Best practices and patterns

HANDS-ON LABS

- → Create reusable modules
- → Multi-environment setup
- → Provision complete VPC
- → Deploy EKS cluster with Terraform

Brief Ansible Overview: We touch on Ansible basics for configuration management. For enterprise Ansible expertise, take our **Advanced Ansible Automation & IaC** course.

12 GitOps & Advanced CI/CD

TOPICS COVERED

- → GitOps principles and workflows
- → ArgoCD for declarative deployments
- → Advanced CI/CD patterns
- → Canary and blue-green deployments
- → Progressive delivery strategies
- → Pipeline as code best practices

HANDS-ON LABS

- → Deploy ArgoCD to cluster
- → GitOps application deployment
- → Implement canary releases
- → Automated rollback strategies

13 Monitoring, Logging & Observability

TOPICS COVERED

- → Observability fundamentals
- → Prometheus metrics collection
- → Grafana dashboards and visualization
- → ELK stack for log aggregation
- → Alerting and incident response
- → SLO and SLI monitoring

HANDS-ON LABS

- → Deploy Prometheus and Grafana
- → Create monitoring dashboards
- → Configure alert rules
- → Set up centralized logging

14 Advanced CI/CD Pipeline Engineering

TOPICS COVERED

- → Jenkins pipeline as code (Jenkinsfile)
- → GitHub Actions advanced workflows
- → GitLab CI/CD pipelines
- → Pipeline optimization and caching
- → Parallel execution strategies
- → Pipeline testing and validation

HANDS-ON LABS

- → Build multi-stage Jenkins pipeline
- → Create reusable GitHub Actions
- → Optimize pipeline execution time
- → Implement pipeline testing

15 Advanced Deployment Strategies

TOPICS COVERED

- → Blue-green deployment patterns
- → Canary releases with traffic shifting
- → Feature flags and A/B testing
- → Progressive delivery with Flagger
- → Rollback strategies and automation
- → Deployment risk management

HANDS-ON LABS

- → Implement blue-green deployment
- → Set up canary with Istio/Flagger
- → Configure automated rollbacks
- → A/B testing implementation

16 Service Mesh & Advanced Networking

TOPICS COVERED

- → Service mesh architecture (Istio/Linkerd)
- → Traffic management and routing
- → Service-to-service security (mTLS)
- → Observability with service mesh
- → Circuit breakers and resilience
- → Multi-cluster networking

HANDS-ON LABS

- → Deploy Istio service mesh
- → Configure traffic routing rules
- → Implement mTLS encryption
- → Set up circuit breakers

17 Advanced Observability & SRE Practices

TOPICS COVERED

- → SRE principles and SLO/SLI/SLA
- → Distributed tracing with Jaeger
- → Advanced Prometheus queries (PromQL)
- → Custom Grafana dashboards
- → Error budgets and burn rates
- → On-call and incident management

HANDS-ON LABS

- → Implement distributed tracing
- → Define SLOs and error budgets
- → Build custom Grafana dashboards
- → Set up incident response workflows

18 DevSecOps & Security Automation

TOPICS COVERED

- → DevSecOps principles and shift-left security
- → SAST/DAST scanning in pipelines
- → Container image scanning (Trivy, Clair)
- → Kubernetes security policies (OPA, PSP)
- → Secrets management with Vault
- → Compliance as code

HANDS-ON LABS

- → Integrate security scanning in CI/CD
- → Deploy and configure HashiCorp Vault
- → Implement OPA policies
- → Automate compliance checks

19 Production Optimization & Cost Management

TOPICS COVERED

- → Performance optimization strategies
- → Resource right-sizing and efficiency
- → Cloud cost optimization techniques
- → Cluster autoscaling (HPA, VPA, CA)
- → Disaster recovery and backup strategies
- → High availability patterns

HANDS-ON LABS

- → Implement cluster autoscaling
- → Optimize resource utilization
- → Set up disaster recovery plan
- → Cost analysis and optimization

20 Capstone Project & Production Deployment

CAPSTONE REQUIREMENTS

- → End-to-end platform design
- → Infrastructure as Code implementation
- → Complete CI/CD automation
- → Production-grade monitoring
- → Security and compliance
- → Documentation and presentation

FINAL DELIVERABLES

- → Working production environment
- → Complete source code and IaC
- → Architecture documentation
- → Runbooks and operational guides
- → Project presentation

Study Tips & Success Strategies

→ **Hands-On Everything:** Every concept must be practiced. Build, break, and rebuild repeatedly.

→ **Document Your Work:** Keep detailed notes on commands, configurations, and troubleshooting steps.

→ **Build Real Infrastructure:** Don't just follow tutorials. Deploy actual applications to production environments.

→ **Infrastructure as Code Always:** Never manually configure cloud resources. Always use Terraform or similar tools.

→ **Version Control Everything:** All code, configurations, and infrastructure definitions must be in Git.

→ **Automate First:** If you do something twice, automate it. This is core DevOps thinking.

→ **Monitor and Measure:** Instrument everything. You can't improve what you don't measure.

→ **Security From Day One:** Never commit secrets. Use proper secrets management from the start.

→ **Learn from Failures:** Deliberately break your deployments to understand failure scenarios.

→ **Join Communities:** CNCF Slack, DevOps forums, local meetups - engage with the community.

→ **Stay Current:** DevOps tools evolve rapidly. Follow release notes and community discussions.

→ **Portfolio Building:** Document all projects on GitHub with comprehensive README files.

Career Outcomes

Upon completing this Core Track, you'll be qualified for these roles:

→ **DevOps Engineer:** Design and maintain CI/CD pipelines, infrastructure automation, and deployment systems

→ **Cloud Infrastructure Engineer:** Architect and manage cloud-native infrastructure on AWS/Azure

→ **Kubernetes Administrator:** Manage and optimize Kubernetes clusters in production

→ **Platform Engineer:** Build internal developer platforms and tooling

→ **Site Reliability Engineer (SRE):** Ensure system reliability and performance at scale

Typical Salary Ranges (Nigeria): ₦4,000,000 - ₦12,000,000+ annually
Global Remote Opportunities: $70,000 - $150,000+ USD annually

Certification Preparation

This course prepares you for these industry certifications:

→ **Certified Kubernetes Administrator (CKA)** - CNCF Kubernetes certification

→ **HashiCorp Certified: Terraform Associate** - Official Terraform certification

→ **Docker Certified Associate (DCA)** - Docker containerization certification

→ **AWS Certified DevOps Engineer - Professional** - Advanced AWS DevOps (with additional AWS study)

→ **Microsoft Certified: DevOps Engineer Expert** - Azure DevOps certification (with additional Azure study)

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